Key: Which Way Did It Blow

84-355 Democracy's Data | The election wind map | Instructor copy — do not distribute

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All code is given to students. Grade the interpretation.

Structure. The brief (`wind-map-brief.Rmd`) and the lab (`wind-map.Rmd`) run the same numbered steps — the brief in prose with three D3 figures, the lab with the code exposed. **Students should read the brief first.** Step numbers below refer to the lab.

This session has two halves and the second one has not happened yet. The 2020-to-2024 map is fully built. The 2026 map is a stub with the 2022 baseline already in it, waiting on an election that falls on **Tuesday 3 November 2026**, inside the semester. See "Running the 2026 map live" at the end.

Every number below was run against the built files.

The session in one sentence

A wind map is the most persuasive picture of an election anyone draws, and every arrow rests on a join that reports its successes and never its meanings — so the lab draws the map, then audits the join, and finds an arrow pointing the wrong way in a place where nothing failed.

The findings

```
c(national_arrows      = nrow(us),
  median_county_swing  = round(median(us$swing), 2),
  national_vote_swing  = round(agg(us$votes_dem_24, us$votes_gop_24) -
                               agg(us$votes_dem_20, us$votes_gop_20), 2),
  dc_naive_swing       = round(F("naive_dc_swing"), 2),
  dc_true_swing        = round(F("true_dc_swing"), 2),
  rows_either_way      = as.integer(F("naive_matched")),
  georgia_counties     = nrow(ga),
  georgia_state_swing  = round(F("ga_state_swing"), 2),
  office_gap_same_ballot = round(F("og_state_gap"), 2),
  precinct_join_ceiling  = F("pj_best_pct"))
```

```
#>        national_arrows    median_county_swing    national_vote_swing
#>                3100.00                   3.06                   6.04
#>          dc_naive_swing          dc_true_swing         rows_either_way
#>                  -0.51                   2.30                3104.00
#>        georgia_counties    georgia_state_swing office_gap_same_ballot
#>                 159.00                   2.47                   2.08
#>   precinct_join_ceiling
#>                  90.90
```

- **88.5% of counties moved toward Trump**, median +3.06. The map is unambiguous and that is part of the problem: it looks like it cannot be wrong.
- **The District of Columbia's arrow reverses**, from -0.51 to +2.30, and **both joins return 3,104 rows.**
- **The counties moved +3.06; the country moved +6.04.** One arrow per county is an editorial position.
- **Two offices on the same Georgia ballots differ by 2.08 points** — as large as Georgia's entire real 2020-to-2024 shift of 2.47.
- **The precinct-level join stalls at 90.9%**, and CHATHAM County loses all 86 precincts.

The three things to get said

1. The row count cannot warn you. This is the session. Two joins of the same two files, 3,104 rows either way, and one of them produces an arrow pointing the opposite direction. **Make somebody say out loud what check would have caught it**, and keep pushing until the answer is "read the `county_name` column, not the count." Every instinct students have about data quality — match rate, missingness, row counts — is silent here.

2. Loud failures are the safe ones. Alaska and Connecticut break the join and the rows fall out; you can count what you lost and say so. The District of Columbia does not break anything. **The ranking students will get backwards is that the loud failure is worse.** Ask which of the three they would rather have in a dataset they were about to publish from.

3. One arrow per county is a choice about whose movement is visible. The counties moved +3.06 and the country moved +6.04. Nothing is miscomputed. The map answers "what did the typical county do", which almost nobody reading it thinks is the question. **This is not a caveat; it is what the figure is.**

The line for the board: a join reports whether it matched, never whether it meant anything.

Walking the steps

Step	What it does	Min	Watch for
1 The encoding	Angle and length, declared as constants	6	Have someone read an arrow back off the legend. If they cannot, nothing later lands.

Step	What it does	Min	Watch for
2 The national map	3,100 arrows, 88.5% pointing right	7	Let them enjoy it first. The audit is more effective after they have believed it.
3 Who published this	No federal county-level source exists	6	**Most students assume the government publishes this.** It does not. Say the FEC line explicitly.
4 Audit the units	Alaska, DC, Connecticut	9	Only Connecticut is a real unit change. Do not let the CT story swallow the other two.
5 The silent failure	DC's arrow reverses; row counts identical	**12**	**The pivot.** Take a prediction before revealing the harmonised number.
6 What harmonising cost	Two of three repairs are deletions	4	A counted hole beats an invented arrow.
7 One arrow per county	3.06 vs 6.04	9	Not a caveat. The figure *is* the choice.
8 Georgia, officially sourced	Same encoding, defensible provenance	6	The only difference from Step 2 is the publisher. Say so plainly.
9 Why not precincts	The join stalls at 90.9%	6	Chatham loses **all** its precincts. The hole would look like quiet.
10 The office problem	2.08 points, same ballots	9	This is what licenses using 2022 rather than 2024 as the 2026 baseline.
11 The 2026 slot	159 tails, no heads	6	Show the stub file on screen. Two empty columns is the whole design.

Step 5 deserves the prediction. Put the two DC rows on the screen — same FIPS, "District of Columbia" and "Ward 1" — and ask: *the join will match this row. Will the resulting arrow be roughly right?* Most of the room says yes, reasoning that DC's wards are all overwhelmingly Democratic so the margin will be similar. **They are almost right, and that is the trap**: the naive margin is -89.4 against a true -86.6, close enough to look fine, and the *swing* built from it points the wrong way.

If you are short on time, protect Steps 5, 7 and 10. The silent join, the editorial weighting, and the measured office effect are the argument. Steps 1 and 2 can be compressed to a demonstration.

The mistake this lab made, and how to teach it

Tell this, because it happened while building the lab.

The first plan was a Georgia **precinct** map — seventeen times the resolution, with a companion lab having already solved the boundary problem by crosswalking 2020's votes onto 2024's precinct lines. All that remained was attaching the 2024 returns by precinct name.

```
read.csv(file.path(D, "precinct_join_ladder.csv"))
```

Rung	Matched	Unmatched 2020	Pct of 2020
1. exact name, uppercased	2198	494	81.6
2. strip leading ballot-order number	2353	339	87.4
3. strip trailing (CITY) tag	2354	338	87.4
4. strip all punctuation and spaces	2446	246	90.9

Each rung is a reasonable repair. Together they reach 90.9% and stop. And the residual is not scattered:

```
head(read.csv(file.path(D, "precinct_join_residual.csv")), 5)
```

County	Unmatched	Total 2020	Ballots 2020	Ballots unmatched
CHATHAM	86	86	131490	131490
COLUMBIA	17	46	79248	31614
DOUGLAS	12	25	68272	33624
CARROLL	11	28	53702	23963
LOWNDES	11	18	45810	17538

CHATHAM County — Savannah — loses every one of its 86 precincts, because the state's returns identify precincts by codes like `1-05C` and the boundary file identifies them by the name of the polling place. There is no regex for that.

The pedagogy: **ask what they would have tried next.** Fuzzy matching and a hand-built alias table are the usual answers, and both are more work in service of a map that would have had an invisible hole where a majority-Black city is. The decision was to drop to county level, where nothing is lost. **Resolution you cannot trust everywhere is not resolution**, and this is the same lesson `precinct-geography` reaches from the other direction.

Handling the room

The figure is genuinely beautiful and that is a teaching hazard.

- **Let them like it.** Five minutes of just looking at Step 2 buys the audit.
- **The audit is not debunking.** The map is basically right. The point is that nothing in the process would have told you if it were not.
- **Nobody writes code.** Say so at the top; the D3 in the brief makes some students assume otherwise.
- **The 2026 half makes this the one session with a sequel.** Flag on day one that they will redraw this map with data that does not exist yet.

Option A — Weight the map by votes

Verified output

```
big <- O[1:100]; sml <- O[(length(O) - 999):length(O)]
data.frame(
  group = c("largest 100 counties", "smallest 1,000 counties"),
  share_of_votes = round(c(100*sum(V[big])/sum(V), 100*sum(V[sml])/sum(V)), 1),
  median_swing = round(c(median(us$swing[big]), median(us$swing[sml])), 2),
  arrows_on_the_map = c(100, 1000))
```

Group	Share of votes	Median swing	Arrows on the map
largest 100 counties	40.1	5.62	100
smallest 1,000 counties	2.3	3.13	1000

```
round(agg(us$votes_dem_24[big], us$votes_gop_24[big]) -
      agg(us$votes_dem_20[big], us$votes_gop_20[big]), 2)
```

```
#> [1] 7.44
```

3/4: Reports that big and small counties moved differently and that the map gives them equal weight.

4/4: Puts numbers on it — the largest 100 counties cast about 40% of the vote and swung a median of +5.6, the smallest 1,000 cast 2.3% and swung +3.1, and the map draws ten times as many arrows for the second group. Concludes that the national swing of +6.0 is invisible on a map whose median arrow is +3.1.

4/4, best: Notices that **there is no weighting that makes the map correct**, because a wind map's job is to show geography and a vote-weighted map is a cartogram that has stopped being a map. Says which question each version answers and picks one on purpose. A student who proposes drawing both and labelling them has understood the figure better than the figure does.

Option B — Break the harmonisation on purpose

Verified output

```
t20 <- table(substr(c20$county_fips, 1, 2))
t24 <- table(substr(c24$county_fips, 1, 2))
k <- union(names(t20), names(t24))
d <- data.frame(state_fips = k, rows_2020 = as.integer(t20[k]),
                rows_2024 = as.integer(t24[k]))
d[!is.na(d$rows_2020) & !is.na(d$rows_2024) & d$rows_2020 != d$rows_2024, ]
```

State fips	Rows 2020	Rows 2024
09	8	9
11	1	8

```r
# states whose row COUNT is unchanged but whose codes all changed
same <- d$state_fips[d$rows_2020 == d$rows_2024]
z <- sapply(same, function(s)
  length(setdiff(c20$county_fips[substr(c20$county_fips,1,2) == s],
                 c24$county_fips[substr(c24$county_fips,1,2) == s])))
z[z > 0]
```

```
#> 02
#> 40
```

3/4: Finds the two states whose row counts differ — Connecticut and the District of Columbia — and describes them correctly.

4/4: Notices that **Alaska has exactly 40 rows in both years and every single code is different**, so a row-count diff misses it completely. States the general rule: comparing counts detects changes in cardinality, not changes in identity.

4/4, best: Turns it round and asks what a state that changed its units *without* changing its count would look like — which is Alaska — and then points out that a state which renamed its units while keeping both count and codes would be invisible to every check in this lab. **Recognising that the audit has a floor is the most sophisticated response available.**

Option C — Audit Georgia against the national file

Verified output

```r
n24 <- c24[substr(c24$county_fips, 1, 2) == "13", ]
n20 <- c20[substr(c20$county_fips, 1, 2) == "13", ]
m <- merge(ga[, c("fips","county","dem_20","gop_20","dem_24","gop_24","swing")],
           n24[, c("county_fips","votes_dem","votes_gop")],
           by.x = "fips", by.y = "county_fips")
m <- merge(m, n20[, c("county_fips","votes_dem","votes_gop")],
           by.x = "fips", by.y = "county_fips", suffixes = c("_n24", "_n20"))
mg <- function(d, g) 100 * (g - d) / (g + d)
m$nat_swing <- (mg(m$votes_dem_n24, m$votes_gop_n24) -
                mg(m$votes_dem_n20, m$votes_gop_n20))
c(counties            = nrow(m),
  gop_2024_difference = sum(m$votes_gop_n24) - sum(m$gop_24),
  dem_2024_difference = sum(m$votes_dem_n24) - sum(m$dem_24),
  dem_2020_difference = sum(m$votes_dem_n20) - sum(m$dem_20),
  max_swing_gap       = round(max(abs(m$nat_swing - m$swing)), 3),
  median_swing_gap    = round(median(abs(m$nat_swing - m$swing)), 4),
  arrows_that_flip    = sum(sign(m$nat_swing) != sign(m$swing)))
```

```
#>            counties gop_2024_difference dem_2024_difference dem_2020_difference
#>            159.0000              7.0000             11.0000           -874.0000
#>       max_swing_gap     median_swing_gap     arrows_that_flip
#>              0.5800              0.0059              0.0000
```

3/4: Reports that the two sources agree closely and concludes the national file is fine.

4/4: Quantifies it — totals differ by a handful of votes out of five million, the median county's swing differs by six thousandths of a point, and **no arrow changes direction** — and then draws the right conclusion, which is *not* that the national file is fine. It is that the file is **accurate here**, and that this could only be established because Georgia happens to publish a check.

4/4, best: States the actual problem: **for 49 other states there is no comparable check available at this cost**, so the correct summary is "verified in one state, unverifiable in the rest" rather than "verified." A student who observes that a single-state audit passing is weak evidence about the other fifty — and that the states most likely to be scraped badly are the ones least likely to publish conveniently — has the argument.

Option D — Argue with the encoding

Verified output

```
ga$turnout_change <- round(100 * (ga$tot_24 - ga$tot_20) / ga$tot_20, 2)
c(state_total_2020 = sum(ga$tot_20),
  state_total_2024 = sum(ga$tot_24),
  state_change_pct = round(100*(sum(ga$tot_24)-sum(ga$tot_20))/sum(ga$tot_20), 2),
  median_county    = round(median(ga$turnout_change), 2),
  counties_losing_votes = sum(ga$turnout_change < 0),
  correlation_with_margin_swing = round(cor(ga$turnout_change, ga$swing), 3))
```

```
#>              state_total_2020              state_total_2024
#>                   4998482.000                   5250048.000
#>              state_change_pct                 median_county
#>                         5.030                         4.340
#>         counties_losing_votes correlation_with_margin_swing
#>                        31.000                        -0.439
```

```
ga[c(which.min(ga$turnout_change), which.max(ga$turnout_change)),
   c("county", "turnout_change", "swing")]
```

County	Turnout change	Swing
Calhoun	-6.29	3.114042
Jackson	25.41	-3.437609

3/4: Builds the turnout map and observes that it looks nothing like the margin map.

4/4: Notes the negative correlation of about -0.44 — counties where turnout grew swung *less* toward Trump — and says what that implies: the margin map and the turnout map are telling two halves of one story and neither is complete alone.

4/4, best: Confronts the identification problem head on. **A margin swing cannot distinguish persuasion from differential turnout**, and the two have completely different implications for what a campaign should do next. Says that a front page needs the margin map and a campaign needs both, and that the wind map is popular partly *because* it collapses a distinction its readers do not know is there.

Option E — Your own question

No verified output. Grade the question as much as the answer. **A 4 requires the question to be answerable from the files given and the student to say what would have changed their mind.**

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is given.
- **The move that earns a 4** is treating an encoding or a join as a decision with consequences that can be measured, and then measuring it.
- **A 2** is "the data has problems so the map is unreliable" — true, useless, and precisely what the lab is built to move past. The map is mostly *right*; the point is that nothing in the process would have said so.
- **Also a 2**: reporting the national numbers as though a government published them.
- **Reward anyone who asks for an error bar on an arrow.** There isn't one, there is no agreed way to draw one, and noticing the absence is the most sophisticated available response.

Anticipated questions

- *"Why is Alaska missing?"* — Because Alaska does not report presidential votes by county, both files invent their own codes for State House Districts, and the districts were redrawn in between. Three separate reasons, any one of which is sufficient.
- *"Couldn't we just fix Connecticut with a crosswalk?"* — In principle, with town-level returns, which Connecticut does publish. It is a real project, not a line of code, and it is a good final-paper topic.
- *"Isn't the map basically right?"* — Yes. Say so. Then ask how they know, and watch the answer turn out to be Option C, which covers one state.
- *"Why not use the FEC data?"* — Because the FEC publishes by state and congressional district. There is no county sheet. This surprises people every time.
- *"Who decides that up means no change?"* — Whoever draws it. There is no standard. Some published wind maps put no-change at horizontal instead, which changes what the map looks like without changing a number.

Connections

- `data-sources` — the same two county files, and the lab that establishes their provenance. **Run that one first if you can**; this lab depends on students already distrusting the file.
- `precinct-geography` — boundaries moving between elections. This lab is the other failure mode: the boundary held and the *label* moved.

- **areal-units** — the unit changes the answer. Step 7 is that lab's point rendered as a picture.
- **migration** — the other arrow map in the corpus, and the reason the length scale here is a declared constant with a printed legend.
- **regional-shift** — the aggregate version of the same question, and a useful contrast: it cannot be wrong in this way because it has no join.

Running the 2026 map live

What to have ready before class

```r
st <- read.csv(file.path(D, "wind_input_2026.csv"),
               colClasses = c(unit_id = "character"))
c(rows = nrow(st), filled = sum(!is.na(st$votes_dem_to)))
```

```
#>   rows filled
#>    159      0
```

```r
names(st)
```

```
#>  [1] "unit_id"        "unit_name"      "state"          "office"
#>  [5] "year_from"      "year_to"        "votes_dem_from" "votes_gop_from"
#>  [9] "votes_dem_to"   "votes_gop_to"
```

- **The baseline is already built.** `wind_ga_counties.csv` holds all 159 Georgia counties with their 2022 governor margins; November 8, 2022 - General/Special Election, snapshot 2025-01-08.

- **One file changes**: `data/wind_input_2026.csv`, two empty columns.

- **One command runs**:

  ```
  Rscript -e 'source("data/build-data.R"); build_2026()'
  ```

  It fetches `export-2026NovGen.json` from `results.sos.ga.gov`, fills the two columns, recomputes every swing, and rewrites the baseline file. Re-knit and the map appears.

The office decision, and why it is not negotiable

The class map is **2022 governor against 2026 governor**, not 2024 president against 2026 governor. Step 10 measures why: two offices on the *same Georgia ballots* in November 2020 differed by 2.08 points, which is the size of Georgia's entire real shift between two presidential elections — and that is with the electorate held exactly constant. A presidential-to-midterm arrow adds a turnout change of roughly a third on top of a floor that already exceeds the signal.

If a student proposes the 2024 comparison, do not just refuse it. Have them compute what it would show, then put the two side by side. The version of this lesson that sticks is the one where they draw the bad arrow themselves.

What will break

Failure	Likelihood	What to do
`export-2026NovGen.json` 404s	**Moderate.** The URL pattern is inferred from 2022 and 2024, both of which return HTTP 200. The state may rename it.	Look at https://results.sos.ga.gov/ in a browser, find the export link, pass it: `build_2026(url = "...")`.
The export exists but has no `localResults`	Low	2022 and 2024 both have it. If not, county totals are on the results page; fill the two columns by hand and call `build_2026(from_stub = TRUE)`.
The contest is not called `Governor`	Moderate	2022 called it exactly `Governor`. Pass `office = "^Governor"` or whatever the export uses.
Counties still reporting	**Certain, if you run this early**	`build_2026()` prints how many of 159 counties have a result. Draw it anyway and label it.
Nothing is posted at all	Low by December	Fall back to the 2022 baseline figure in Step 11 and run the session as a design exercise: *what would you need to see before you drew the arrow?* That is a legitimate 80 minutes.

If the data is not out

Do not cancel the step. Run it as a pre-registration exercise: have the room predict the 2026 map before it exists — direction of the statewide arrow, which counties move most, whether metro Atlanta and rural Georgia move together — and write the predictions down. Then redraw it when the data lands, in office hours or by email. **A prediction recorded before the data arrives is worth more than the map**, and it is the only exercise in this course that can do that, because it is the only one whose data has not happened yet.

A note on the figures

The interactive (D3) and printed (base-R) versions of every map read the same `x`, `y`, `dx`, `dy` columns out of the same CSV. Neither one projects or rescales anything, so they cannot drift. Checked mechanically against the knitted HTML: the two renderers place arrow tips within **1.0 pixel** on the national map (the whole-pixel rounding floor of the D3 payload) and within **0.1 pixel** on the Georgia map, on figures 780 and 720 pixels wide.

```
read.csv(file.path(D, "parity.csv"))
```

Figure	N	Max dx err	Max dy err	Max len err
national counties	3100	0	0	0
Georgia counties	159	0	0	0

The provisional label

Whatever gets drawn in early December will be **unofficial**. Certification runs into December in many states, close races will still be open, and precinct-level returns will not exist. Put that on the figure, in words, on the slide. **Every wind map published on election night has this property and almost none of them say so**, which is itself the last thing this lab has to teach.